VF Asia Footwear Laboratory Testing Manual

Test Method for Ink Adhesion on Substrate	Test method # VF- F05 REF TM-R- 038 Page 1 of 2
Date Issued : July 11, 2014 Revision # : 1 Reason for revision: Follow TBL rating system for failure.	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To test the adhesion of ink that is applied on substrate

REFERENCES: None

EQUIPMENT AND MATERIAL:

1. Scotch tape 3M # 600 or 3M # 810 (TBL requirement)
2. Hand press-roller or 1-2 kg load.
3. Cutting devise such as a sharp knife or a pair of scissors
4. Distilled water
5. Immersion dish which is large enough to contain the test sample
6. Sodium Chloride (NaCl) – AR Grade
7. All types of material with coating and print.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Prepare the test sample which was printed for at least 24 hours
3. Cut specimen with at least 2 cm width (bigger sample is more ideal
4. Prepare a 3% sodium chloride water solution. To prepare 3%, weigh 3 grams of NaCl , dissolve in 97 ml of distilled water.

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A. Dry Test :

1. Lay the sample flat on the table.
2. Remove any dust or loose contaminants that may affect the surface contact between the printed sample and the scotch tape.
3. Press a strip of scotch tape 3M # 600 firmly on the printed surface of the sample and making sure the tape completely covered the required area , keeping one end of the tape free (not adhered to the surface).
4. Roll the hand press on top of the scotch tape to ensure that the adhesion of the scotch tape is good, and wrinkle - free.

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5. Hold the free end of the tape at a 90 – degree angle to the sample and separate with one quick pulling motion.
6. Rate the leather on a scale of 1 to 3 as follows:

Rating	Description
1	No visible finish (ink/paint) removal
2	Slight finish (ink/paint) removal that is restored when rubbed with finger
3	Permanent finish (ink/paint) removal that is not restored when rubbed with finger

B) Wet Test – Distilled Water.

1. Soak the prepared test samples in distilled water at 23.0+/-1.0 ºC for 2 hours.
2. Remove the sample from the water, and test immediately following the directions outlined in "A) Dry test "above.

C) Wet Test – Artificial Sea Water Solution:

1. Soak the prepared test samples in artificial sea water solution at 23.0+/-1.0 ºC for 2 hours.
2. Remove the sample from the artificial sea water solution, and test immediately following the directions outlined in "A) Dry test "above.

REPORT: Rating at the scale 1-3.